HandRacer: A gesture-based racing game

1, Project Overview

HandRacer is a gesture-controlled racing game based on Leap Motion. Players use their hands to simulate the posture of holding a steering wheel to control the vehicle's steering. Throttle and brake are controlled by pressing the left palm up and down, while a right-hand push gesture triggers power-up abilities. The game aims to deliver an immersive driving experience, turning the player's own body into the controller.

2, Game Modes

Online Battle	**Player vs Player (PVP)**	Two players compete in real time. Player 1 uses keyboard. Player 2 uses Leap Motion gesture control. Race head-to-head!
AI Battle	**Player vs Computer (PVE)**	Single player versus an AI opponent. Choose either keyboard or Leap Motion gesture control to race against the computer.

3, Keyboard Controls

For: Keyboard players (Online Mode — Player 1 / AI Mode)

Key	Function
`W`	Accelerate — move forward
`S`	Brake / Reverse
`A`	Turn Left
`D`	Turn Right
`SPACE`	Pick up item / Use item

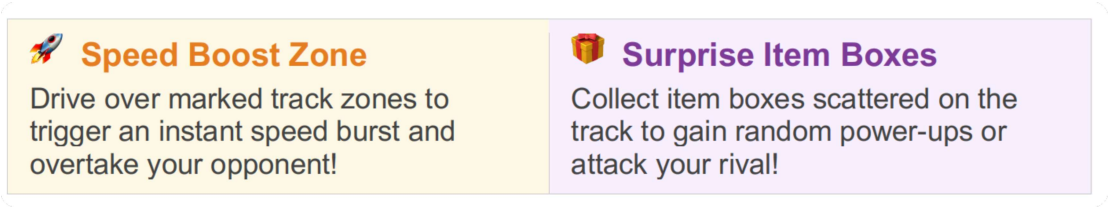
4, Leap Motion Gesture Controls

For: Gesture players (Online Mode — Player 2 / AI Mode)

Keep your hands 25–60 cm above the Leap Motion sensor for best detection

Left hand — Press Down	Accelerate — lower = faster
Left hand — Rotate Left/Right	Steer — rotation angle = steering angle
Right hand — Push Forward	Activate & use current item

5, Track Features



6, Highlights and Innovations

Frontier Interaction&Architecture

Intuitive natural interaction (NUI) design: Deeply optimized gesture mapping for Leap Motion. Lower the control throttle with your left hand, rotate the simulated steering wheel, and release the prop by pushing the palm with your right hand. This set of operational logic is highly compatible with real driving intuition and ergonomics, and can be instantly understood by both operators and observers, achieving a truly "zero threshold" immersive experience.

Flexible dual ended control and low coupling architecture: Provides a complete WASD keyboard and Leap Motion gesture dual input solution. Thanks to the decoupled input management design at the bottom, seamless switching of control modes can be achieved by simply adding the - leave suffix at startup, greatly improving the deployment flexibility of the program and leaving interfaces for future integration of more interactive hardware.

Efficient performance optimization design: The game adopts Low Poly stylized scenes, which not only form a unique artistic expression, but also effectively control the number of faces and rendering costs of the model from the bottom. Cooperate with possible resource management strategies (such as prop object pool mechanism) to ensure high frame rate operation even in multiplayer and complex physical collisions.

Gameplay and Balance

4. Dynamic track and asymmetric competitive balance: Innovatively introducing dynamic track mechanisms (such as unlocking new routes and shortcuts in the second lap). This not only breaks the repetitive feeling of traditional racing laps, but also cleverly balances the accuracy differences between different input terminals of "gesture recognition devices" and "traditional keyboards" in level design, ensuring fairness in cross end competition.

5. Diversified game strategy prop system: Deeply integrated with prop battle gameplay, players can pick up various props such as acceleration, banana peels, shields, etc. in the track. This has changed the purely competitive racing mode, introducing randomness and strategic games, greatly enhancing the fun and program effect of friend battles.

6. Intelligent AI and Online Social Ecology: Supports highly intelligent human-machine battles and LAN/online multiplayer connections. The game is equipped with a real-time updated post game ranking system, which effectively drives players to repeatedly challenge through a competitive mechanism, significantly improving the game's replay rate and social attributes.

Audiovisual Presentation and Engineering Completion (Audio Visual&Polish)

7. Extremely tense stylized audio-visual experience: perfectly integrating high-speed driving with stylized art. Through carefully tuned lens FOV changes, realistic velocity lines, and drift particle effects, it brings a strong visual impact. At the same time, the game is equipped with exquisite original posters and dynamic background music, providing a highly unified audio-visual experience.
8. Commercial grade quality interactive packaging (UI/UX): with engineering completion of mature games. Featuring high-quality main menu UI navigation logic, supporting comprehensive lifecycle management such as global call out pause for Esc, clear interface feedback, and providing players with a smooth and complete gaming experience loop.

Ps: How to switch leaf motion mode from wasd control: Write “C: \ Games. exe (space)-leap” in the shortcut or command line.

7, Team Responsibilities

1) Conceptualized the innovative topic: All team members
2) Unity setup and overall model design: Liu Yuehan, Zhou Tianjian
3) Leap Motion integration and design: Hu Zeyun, Jia Qiancheng
4) UI interface design: Liu Yuehan, Zhou Tianjian
5) Program debugging and testing (later stage): All team members
6) Demo video filming and summary document writing: All team members

